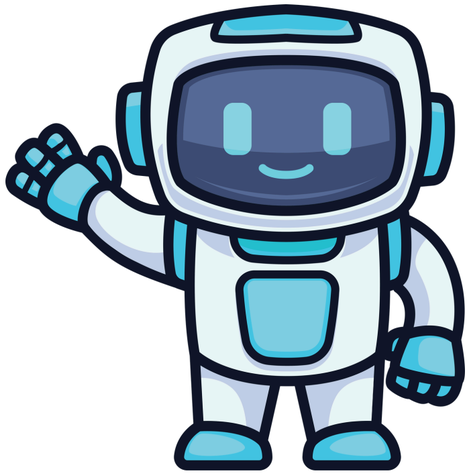
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(Autonomous)



CS5304-MACHINE LEARNING

Presentation by

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TITLE OF THE PROJECT

MindTrace AI:
A Cognitive Integrity Preservation System (CIPS)

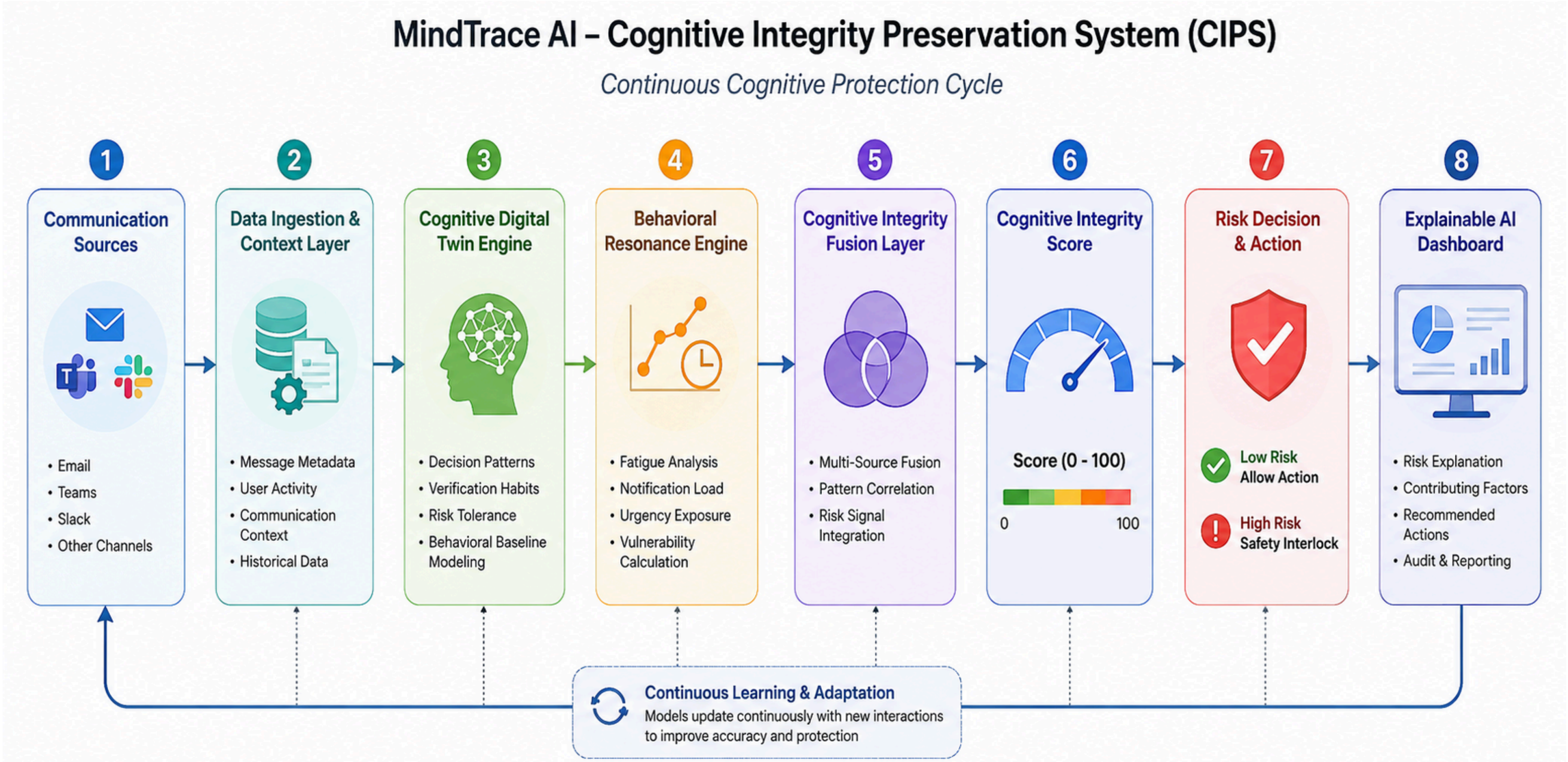
SUPERVISION

Dr.Aruchana

CONTENTS

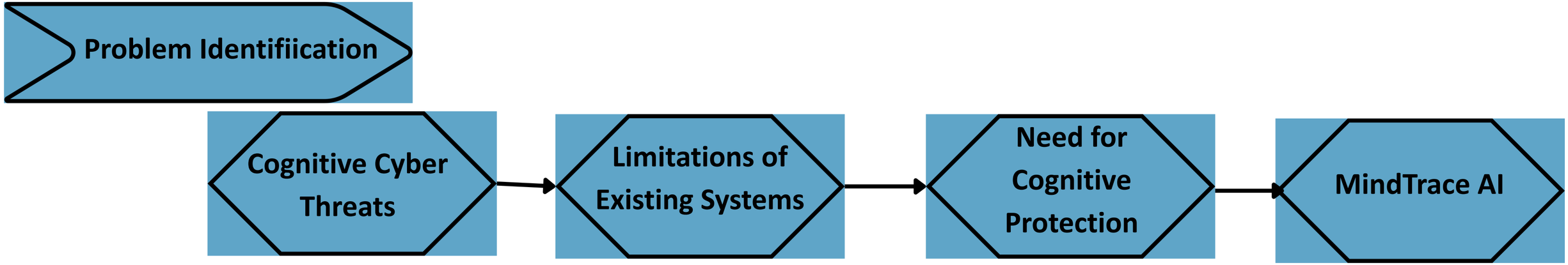
1	INTRODUCTION TO PROBLEM STATEMENT
2	PROBLEM STATEMENT AND ITS DESCRIPTION
3	PROJECT DESCRIPTION AMD ITS KEY FEATURES
4	LITERATURE SURVEY
5	TECHNICAL APPROACH TO PROPOSED SOLUTION
6	AIM AND SCOPE OF THE PROPOSED SOLUTION

I.Introduction (1/1)



II.Problem Statement (1/1)

Existing cybersecurity solutions protect digital assets but fail to safeguard users from cognitive manipulation, misinformation, and psychological cyber threats. MindTrace AI aims to preserve cognitive integrity by using AI to detect behavioral risks, provide early warnings, and enhance safe digital decision-making. The system continuously analyzes user behavior and digital interactions to identify patterns associated with cognitive risks. By integrating artificial intelligence with behavioral analysis, MindTrace AI promotes digital well-being, strengthens cyber resilience, and supports informed decision-making in an increasingly complex online environment.



III.PROJECT DESCRIPTION AND ITS KEY FEATURES(1/1)

1. DATA COLLECTION & INPUT



Collects digital behavior and interaction data from various online sources:

- Browsing activity
- Social media interactions
- Communication patterns
- App usage and screen time
- Device and network information

2. MINDTRACE AI ANALYSIS ENGINE



AI models analyze data to understand cognitive and behavioral patterns:

- Behavioral analysis
- Emotion and sentiment detection
- Cognitive risk assessment
- Misinformation & manipulation detection
- Social engineering & phishing detection

3. OUTPUT & COGNITIVE INTEGRITY PRESERVATION



Generates insights and takes action to preserve cognitive integrity:

- Risk score and threat alerts
- Personalized recommendations
- Digital well-being monitoring
- Awareness and safety guidance
- Cognitive integrity protection

EXISTING SYSTEM



- Current cybersecurity systems focus on protecting devices, networks, and data from external threats.
- Mental health and cognitive well-being are managed separately through traditional assessments or self-reporting methods.
- No real-time monitoring of cognitive manipulation, misinformation, digital addiction, or online psychological influence.
- Lack of integration between AI, cybersecurity, and cognitive health monitoring.
- Existing solutions are reactive and do not prevent cognitive risks in digital environments.

PROBLEMS IN EXISTING SYSTEM



- Inability to detect cognitive manipulation and psychological attacks in real time.
- Lack of personalized monitoring of users' cognitive and emotional well-being.
- Security solutions ignore the human cognitive aspect of cyber threats.
- Existing mental health tools are reactive rather than preventive.
- No integrated system to preserve cognitive integrity against digital misinformation, phishing, social engineering, and AI-generated deceptive content.

PROPOSED SOLUTION



- Develop MindTrace AI, a Cognitive Integrity Preservation System (CIPS) that uses AI to monitor, analyze, and protect users' cognitive well-being in digital environments.
- Detect patterns of misinformation, social engineering, emotional manipulation, addictive behavior, and suspicious online interactions.
- AI-based behavioral analysis provides early warnings and personalized recommendations to reduce cognitive risks.
- Combine cybersecurity, behavioral psychology, and AI to preserve cognitive integrity, enhance digital resilience, and promote safe online decision-making.

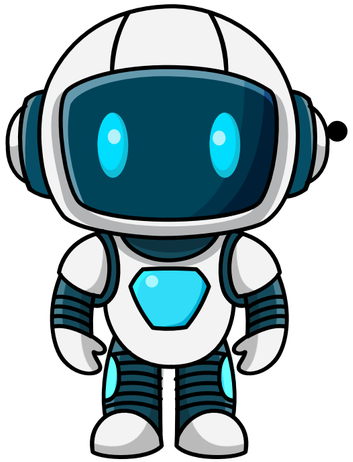
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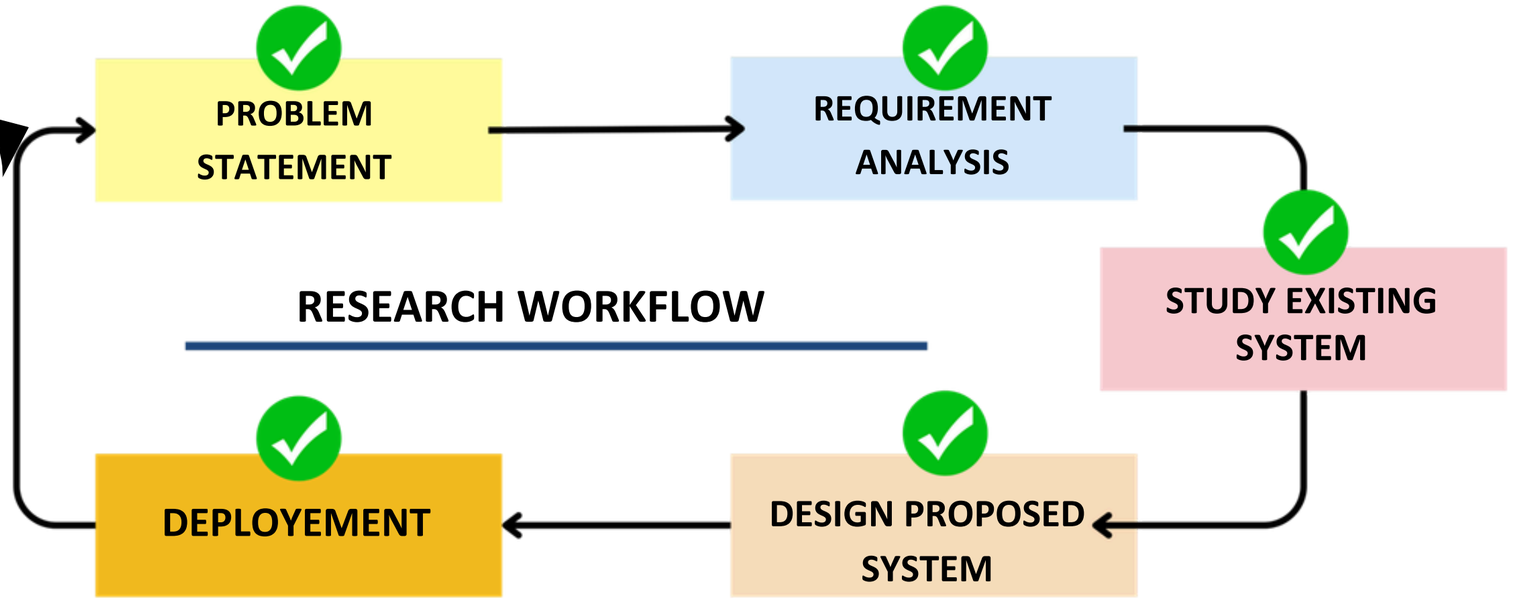
LITERATURE SURVEY

SR. NO.		RESEARCH PAPER TITLE	AUTHOR NAME	DATE & YEAR	METHOD / FEATURE EXTRACTION	LIMITATIONS	PREPROCESSING
1		Forensic Cyberpsychology and the Human Side of Cybercrime: The Mind Behind the Screen	J. McDowell	2026	• Theoretical Framework of Forensic Cyberpsychology • Psychological Trait Analysis • Affective Computing Features • Behavior Pattern Recognition	• Conceptual in nature • Lack of large-scale empirical validation • Limited dataset availability	• Text Normalization • Stop Word Removal • Tokenization • Sentiment Labeling
2		Exploring the Complex Nature of Cyberpsychology Risks to Children and Beyond the Screen	R.K. Gorea	2025	• Risk Factor Identification • Behavioral Analysis • Digital Interaction Features • Victim Profiling Model	• Focus only on children population • Limited cross-platform data • Context-specific results	• Data Cleaning • Handling Missing Values • Encoding Categorical Variables • Text Preprocessing
3		Beyond the Screen: The Expanding Role of Forensic Nursing in Managing Cybercrime Victimisation	R.K. Gorea	2025	• Forensic Nursing Intervention Framework • Psychological Impact Analysis • Victim Support Indicators • Case Management Features	• Limited technological focus • Smaller sample size • Domain-specific applicability	• Data Cleaning • Standardization • Missing Value Imputation • Encoding & Scaling
4		Forensic Perspectives on Cybercrime: Human Behavior and Cybersecurity	John McAlaney, Peter J. Hills, Terri Cole	2024	• Human Behavior Modeling • Cybersecurity Analytics • Digital Footprint Features • Threat Prediction Framework	• Limited real-time datasets • Model generalization challenges • Privacy and ethical issues	• Data Normalization • Feature Scaling • Outlier Removal • Label Encoding

IV.Aim & Scope of the Proposed Work (1/1)

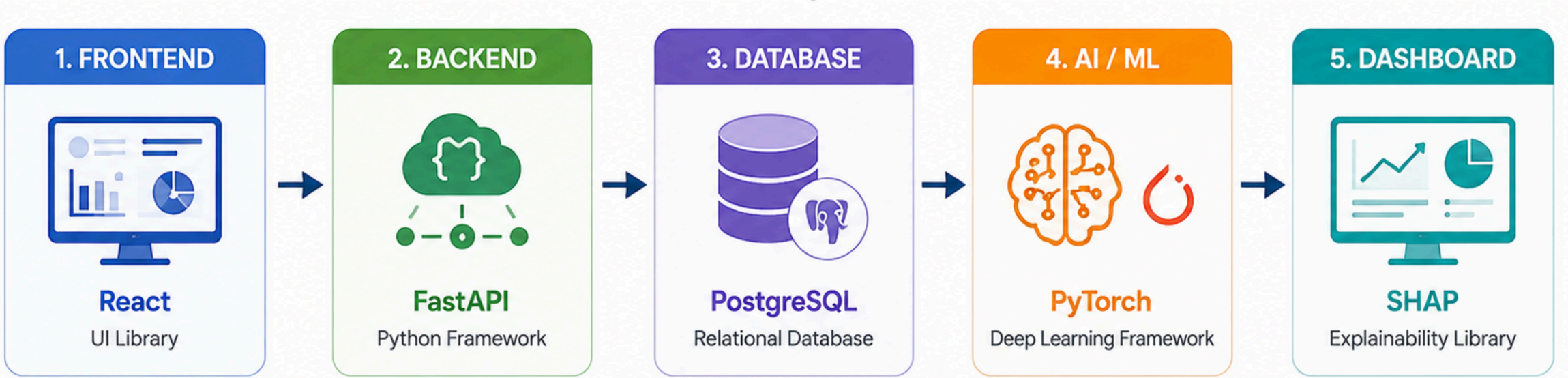
- Cognitive Threat Detection
- Cognitive Integrity Preservation
- AI-Based Behavioral Analysis
- Personalized Risk Assessment
- Digital Resilience





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TECHNICAL APPROACH TO THE PROPOSED SOLUTION



COMPONENTS OVERVIEW

Frontend (React)
Provides an intuitive user interface for monitoring cognitive integrity, risk scores, alerts, and insights.

Backend (FastAPI)
Handles API requests, business logic, user authentication, data processing, and integration with AI services.

Database (PostgreSQL)
Stores user profiles, communication metadata, behavioral patterns, risk scores, and system logs securely.

AI / ML (PyTorch)
Powers cognitive digital twin, behavioral resonance analysis, risk scoring, and vulnerability prediction using advanced ML models.

Dashboard (SHAP)
Provides explainable AI insights by showing the factors contributing to each risk score and recommended actions.

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THANK YOU !!

